Developmental Psychology
1970, Vol. 3, No. 3, 298–306

Mother-Infant Separation in the Patas Monkey[1]

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Studies of human beings and several macaque monkey species have indicated a similar reaction to maternal separation, involving a short-lived agitation, followed by a period of physical and mental depression, and, in human beings, sometimes culminating in social indifference and sociopathic behavior. It was found that the six 7-month-old patas monkey infants used in this study showed the typical initial agitation and subsequent decline in social behavior during the 3-week separation; but that an increase in peer interaction, rather than the common rise in maternal interaction found in other species, followed reunion. Three factors may interact to produce a less severe separation reaction and increased peer interaction following separation, as compared to the results of macaque studies: (*a*) patas tend to be more permissive mothers, (*b*) patas infants have a longer, more intense, early mother-infant relationship, and (*c*) contact play is a secondary play pattern in patas, but is the primary play pattern in macaques.

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Bowlby (1958, 1960a, 1960b, 1961) advanced an ethologically based theory accounting for mother-infant attachment and the consequences of maternal separation. He divided the child's reaction to separation from his mother into stages of protest, despair, and detachment; the psychological correlates of which are separation anxiety, grief and mourning, and defense reactions. Bowlby conceptualized this sequence as resulting from the removal of the object of the several "component instinctual responses" (sucking, following, clinging, crying, smiling) which comprise the child's tie to his mother.

Spitz (1954) saw the course of infant development in a bleak foundling home as an adaption to an extremely stressful situation, similar to the General Adaptation Syndrome (Selye & Fortier, 1950). The mental and physical retardation Spitz (1946) called "anaclitic depression" was reversible if maternal separation lasted less than 3 months. Separation of more than 5 months resulted in an irreversible depression, frequently ending in death, which Spitz (1945) labeled "hospitalism."

However, Pinneau's (1955) detailed review strongly suggested that maternal separation was, at best, of minor importance in Spitz's studies. This and subsequent investigations (e.g., Caldwell, Hersher, Lipton, Richmond, Stern, Eddy, Drachman, & Rockman, 1963; Gardner, Hawkes, & Burchinal, 1961; Schaffer & Emerson, 1964; Stott, 1962; Yarrow, 1964) have made it evident that Bowlby's and Spitz's rather unequivocal conclusions concerning the effects of maternal deprivation were based on the effects of many confounded variables. Stott (1962) in particular presented a strong case for the organic bases of much of what had been conceived of as psychological damage.

A "retrospective follow-up" study (Bowlby, Ainsworth, Boston, & Rosenbluth, 1956) in which adolescents were first studied some years *after* they had been isolated in hospital for infantile tuberculosis,

[1] This research was supported in part by National Science Foundation Grant GB 7601, and a grant from the Louisiana State University Graduate Research Council. Subjects were provided by the Delta Regional Primate Research Center, United States Public Health Service Grant No. FR 00164 from the Division Research Facilities and Resources. Thanks are due to Dalton Day for his assistance in analyzing the data.

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illustrates some of the confounded effects involved in maternal separation in children. On the basis of this study Bowlby conceded that reactions to separation are quite variable and the children do not "commonly" develop psychopathic and affectionless characters, as he had first asserted (Bowlby, 1952).

Although maternal separation per se is not of such overwhelming consequence in child development as had been thought, it does provide information relevant to conceptualizations of mother-infant attachment behavior. The closeness of the mother-infant relationship in monkeys and its similarity to that in human beings has prompted several studies of maternal deprivation and separation in macaque monkeys. Harlow's classic studies (e.g., Harlow, 1961; Harlow & Harlow, 1962; Harlow & Zimmerman, 1959) of rhesus (*Macaca mulatta*) infants raised on wire and/or cloth surrogate mothers, or subjected to other forms of maternal deprivation, indicated that "contact comfort," maintained by clinging, was the primary cement bonding the rhesus infant to his mother. As for long-term effects, monkeys raised with no mother or peer experience become asocial adults, while those infants reared with their mothers but no peer experience grew up to be only somewhat less asocial. The mother-infant tie must be complemented by peer experience in order to produce a "normal" laboratory monkey. Maternal separation most clearly affects peer interaction, as indicated by the following studies of monkeys.

The studies of maternal separation in rhesus monkeys (Griffin, 1966; Mitchell, Harlow, Griffin, & Møller, 1967; Seay, Hansen, & Harlow, 1962; Seay & Harlow, 1965; Spencer-Booth & Hinde, 1966) indicate a reaction closely paralleling that seen in children. There is an initial burst of agitated motor activity and intense cooing (crying) vocalizations, followed by a great decrease in behavior generally and peer-directed social behavior in particular. "Detachment" (Bowlby, 1961), wherein a child does not respond to his mother if and when she returns, was not seen in these studies.

Rosenblum and Kaufman (1968) studied maternal separation in a group of bonnet *(Macaca radiata)* and a group of pigtail *(Macaca nemestrina)* macaques. The bonnet infants showed the typical agitation initially, but soon attached themselves to substitute mothers. The presence of substitute mothering apparently prevented the depression seen in rhesus and pigtail monkeys. The pigtail infants' depression was the most severe described in any study of maternal separation in monkeys. These infants sat hunched in a ball, isolated from their surroundings, for nearly a week before they began moving about the pen again. After reunion the mother-infant relationship intensified, and remained so for over 3 months.

Apparently there is considerable variability in response to maternal separation among different monkey species. By extending this research to nonmacaque species it should be possible to further understanding of the nature of species differences and of the underlying dynamics of mother-infant and infant-infant social relationships.

Erythrocebus patas, with which the present research was concerned, is an African, semiterrestrial, troop-living Cercopithecine monkey. Field and laboratory studies have recently provided a picture of the basic social structure of a patas monkey group. Hall (1966) found that the wild patas monkey lives in groups of about 15 members, with only one adult male per group. Patas differ from macaques and baboons in being silent and relatively less aggressive. The adult females control most group behavior, while the male stays apart, watching for predators and particularly for other male patas monkeys. Hall and Mayer (1967) reported that in a laboratory group of patas, the adult male remained aloof from the group but when a juvenile male became pubescent the adult male attacked him viciously and persistently, forcing the juvenile's removal. The play behavior of young patas consists of rapid chasing and bounding off bushes, unlike the wrestling play typical of macaques. Also unlike macaques, young patas play vigorously with their mothers.

Seay, Schlottmann, and Thorne (1970) have found these patterns to hold in a laboratory setting. Throughout the first half-

year of development, patas infants interact much more with their mothers than do macaques. As rejection and threat behavior are higher for macaque mothers than patas mothers, the mother monkey rather than the infant apparently exerts the limiting force in mother-infant interactions.

Rosenblum and Kaufman (1968) related the greater permissiveness of bonnet macaque mothers to greater autonomy in their offspring. However, because each of the separated bonnet infants had a substitute mother who accepted him and the pigtail infants did not, the effects of the quality of preseparation mothering on the reaction to separation cannot be directly assessed in their study.

The present study was designed to permit some tentative conclusions as to how much influence greater mother-infant interaction, but more permissive mothering, has on infant monkey development. Also, because the method was almost identical to that used by Seay et al. (1962), Seay and Harlow (1965), and Seay et al. (1970), close comparisons of macaque and patas reactions to separation were possible.

Method

Subjects

The subjects were six infant patas monkeys and their mothers, with whom the infants had lived from birth. Four of the infants were male. At separation the infants ranged from 203 to 237 days of age.

Apparatus

A basic metal-mesh housing unit consisted of two 36 × 27 × 42 inch home cages at opposite ends of a 60 × 27 × 42 inch play cage which the infants, but not the mothers, could enter via a 3.5 × 5 inch opening in each home cage. The play cage could be divided in half by a vertical metal-mesh panel. One mother-infant dyad was housed in each home cage, thus allowing each infant access to one other mother and infant.

Procedure

The study was divided into three 3-week periods as follows: (*a*) preseparation—base-line data were taken while the infants lived with their mothers; (*b*) separation—the mothers were removed and individually housed in another wing of the laboratory. The infants lived alone during this period except for 1 hour a day when the panel in the play cage was removed; (*c*) reunion—the mothers were returned to their respective home cages and lived with their infants as during preseparation.

Separation was accomplished without force by covering the entrances to the home cages when the infants were in the play cage. The mothers were put into transport cages and removed from the room. The home cage entrances were then reopened and data collection was begun.

Throughout the study the panel dividing the play cage was removed 1 hour a day, 5 days a week, to permit each infant to interact with the other infant. Data were collected during these interaction periods in eight 15-minute sessions a week, using a revised form of Hansen's (1962) symbol checklist. Symbols were written on three dittoed sheets divided into 20 equal sections, which were further halved to separate mother scores from infant scores. A tape-recorded man's voice counting from 0 to 60 divided a 15-minute data collection session into 15-second intervals. Behavior symbols were entered in a new section of the data sheets, corresponding to each timed interval. A subject's score for a given behavior was the number of 15-second intervals in which it was entered in one 15-minute session. The possible score ranged from 0 to 60.

An observer scored the behavior of one mother-infant pair while sitting in view of the subjects, 4 feet away from the cage. The two observers who collected data for the study randomly alternated pairs from session to session.

The interobserver reliability for each behavior category was determined by computing product-moment correlation coefficients on the totaled session scores of data collected simultaneously by the two observers on the same mother-infant pair. The correlations were based on 33 such sessions, collected during the 6 months preceding the study. The categories discussed in this study are given in Table 1. For a more complete list of categories scored see Thorne, Schlottmann, and Seay (1969).

Statistical Analysis

Each subject's mean score per session for each behavior category, for each experimental period, was calculated and a repeated-measures analysis of variance was performed. Infant-infant and infant-nonsocial behavior categories were analyzed across preseparation, separation, and reunion. Post-ANOVA Neuman-Keuls tests (Winer, 1962) were used to determine the particular pattern of significant differences for these categories. All other categories which involved behavior of the mothers were analyzed for differences between preseparation and reunion. Immediate effects of experimental manipulations were evaluated by *t* tests based on single day scores as follows: last preseparation versus first separation day; first separation versus last separation day; last preseparation versus first reunion day. The .05 level of significance was accepted for all tests.

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TABLE 1

BEHAVIOR CATEGORIES

Category	Definition	*r*
	Social play and aggressive behavior	
Noncontact play	Highly active movement toward and/or away from another animal involving two or more directional changes; also includes beeline mock attacks of three or more body lengths and rapid bouncing in place with orientation toward another animal	—[a]
Contact play	Wrestling or biting with rapid changes in location, or biting with headshaking involving another animal	—[a]
Aggression	Intense rapid biting, wrestling or clasp-pulling another animal; often accompanied by piloerection and vocalization; distinguished from contact play by increase rather than cessation of activity when other animal submits	—[a]
Jaw drop	Opening the mouth while orienting to another animal	.85
Approach	Oriented movement of one body length or more toward another animal	.83
Withdraw	Oriented movement of one body length or more away from another animal	.88
Clasp-pull	Clasping and jerking the fur or skin of another animal	—[a]
	Other social behavior	
Nonspecific contact	Brushing against or touching another animal if not subsumed under another category (scored only for infants)	.98
Nonventral gross contact	Gross body contact with another animal's body, excluding the ventral surface (scored only for infants)	.98
Ventral contact	Gross body contact with the ventral surface of another animal (scored only for infants)	.94
Groom	Spreading and picking manually at the fur of another animal	.95
Lipsmack	Rhythmic parting and closing of the lips while oriented to another animal	.87
Nipple contact	Holding the nipple of another animal in the mouth (scored only for infants)	.99
Thrust	Rhythmic, pistonlike pelvic movements against another animal	—[a]
	Nonsocial and miscellaneous behavior	
Coo	Tonal vocalization resulting in "ooooh" sound	.97
Self-play	Same as noncontact play, but with no orientation to another animal	—[a]
Oral manipulation	Any mouthing, sucking, or licking of the cage or other object	.88
Manual manipulation	Fingering or other manual exploration of the cage or other object	.88
Oral self-manipulation	Any mouthing of the animal's own body.	.96
Eating	Mouthing or biting of food pellet, but not chewing	.98
Play cage location	All four feet in the play cage	.99
Other home cage location	All four feet in the other home cage	.99

[a] The behavior was recorded in too few instances to permit a stable estimate of category reliability.

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Results

Immediate Effects

Separation. Immediate reactions of the infants to removal of their mothers (see Table 2) included frequent and intense cooing, frantic searching about in all three sections of the apparatus, and wide-eyed scanning of the room. The full pattern lasted less than half an hour, but the visual searching remained above normal for several days. The infants also tended to stay near each other, as reflected in the higher nonspecific-contact mean score on the first day of separation. Other activities usually engaged in by the infants fell to low levels immediately after removal of the mothers. Nonsocial oral and oral self-manipulation declined significantly. Noncontact play decreased, but not significantly, due in part to the rapid searching and following behavior of the infants, which frequently met the criteria for play, although qualitatively different from the usual noncontact-play pattern.

Comparisons of the behavior of the infants on the first and last days of the separation period generally indicated a return to preseparation levels by the last separation day. Cooing decreased ($p < .05$), while nonsocial oral and manual manipulation and eating increased ($p < .05$). Infant-infant nonspecific contact decreased ($p < .05$), but the decrease was small.

Reunion. In many ways, return of the mother produced immediate but short-term changes in infant behavior corresponding to the immediate reactions to her removal (see Table 2). Cooing increased and oral self-manipulation decreased, both significantly, and nonsocial oral and manual manipulation showed some tendency to decline ($p < .10$). Infant-infant nonspecific contact decreased significantly, comparing last separation day with first reunion day. Immediately on reunion, infant-mother nonventral gross contact and oral manipulation rose significantly, but nipple contact did not. Mother-infant approaches, lipsmacks, and grooming increased significantly, but the increased infant-mother and mother-infant interaction was short-lived. Four subjects began peer play that was qualitatively like that

TABLE 2

COMPARISONS OF ONE-DAY MEAN SCORES

Behavior	Last preseparation-first reunion		Last preseparation-first separation		First separation-last separation	
Infant-infant nonspecific contact	19.75	8.92	19.75	38.25	38.25	30.83
	*		*		*	
Infant-infant noncontact play	6.33	3.50	6.33	3.42	3.42	2.08
Infant coo	6.92	28.95	6.92	40.92	40.92	7.42
	*		*		*	
Infant nonsocial oral manipulation	16.42	5.42	16.42	5.25	5.25	15.18
			*		*	
Infant-mother nonventral gross contact	2.08	11.50				
	*					
Nipple contact	1.18	0.58				
Mother-infant approach	0.0	1.18				
	*					
Mother-infant groom	0.92	6.58				
	*					

* $p < .05$.

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of preseparation within 15 minutes of their mother's return.

Overall Three-Week Effects

Infant-infant behavior. The immediate increase in infant cooing in separation did not continue long enough to produce a significant increase over the whole separation period. Infant oral self-manipulation and self-play decreased ($p < .05$) in separation and remained low in reunion, while nonsocial oral manipulation did not decrease ($p < .05$) until reunion (Figure 1).

Eating by infants increased ($p < .05$) in separation and fell ($p < .05$) to preseparation levels in reunion. Both infant nonspecific contact (Figure 2) and nonventral gross contact increased ($p < .05$) in separation and returned ($p < .05$) to preseparation levels in reunion. Infant noncontact play decreased significantly in separation and rose to the preseparation level in reunion (Figure 2). Clasp-pull and jaw-drop scores for infants increased ($p < .05$) only in reunion (Figure 3).

Contact play was scored for three subjects. In the two male-female infant pairs, each male received one contact-play score in separation. In the male-male pair a total of 11 contact plays were scored in separation and 198 were scored in reunion. Contact play had rarely been observed, formally or informally, in the Louisiana State University laboratory for patas infants prior to this time. Thrusting and aggression were also scored, but infrequently. Nine instances of aggression by three subjects were seen in separation and a fourth subject received one score in reunion. All but one aggression was scored for male infants. There were no other apparent sex differences in behavior among the infants. Approach and withdraw scores followed the pattern of other play components, but did not reach statistical significance. Play-cage-location scores did not differ across the three experimental periods, but other-home-location scores, which were almost nonexistent otherwise, increased to the level of play-cage-location scores in separation.

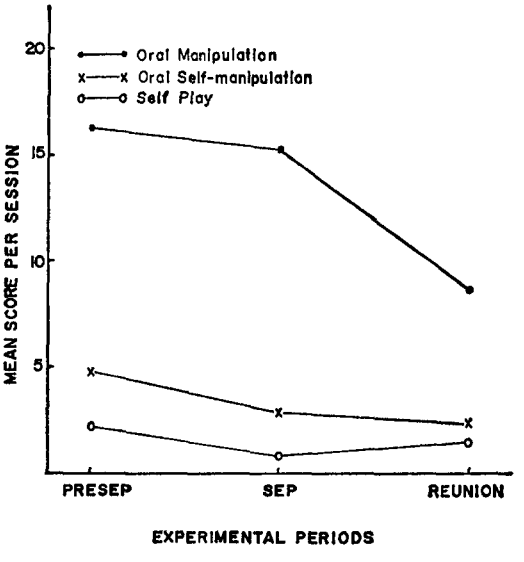


FIG. 1. Infant nonsocial behaviors.

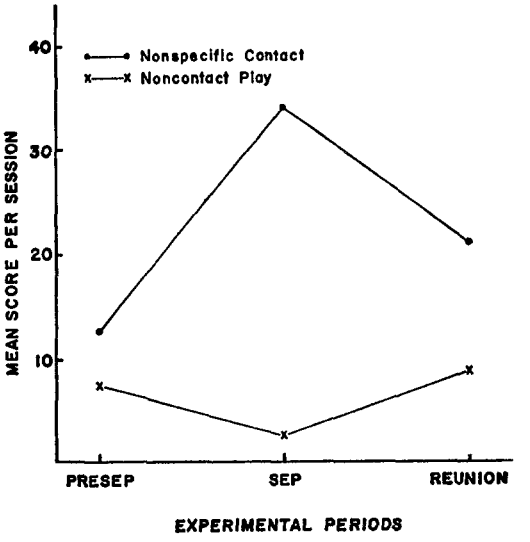


FIG. 2. Infant-infant behaviors.

Other behavior. All mother-directed behaviors of the infant decreased in reunion as compared to preseparation levels, but only approaches declined significantly, from a preseparation mean score per session of 10.57 to 4.28 in reunion. Mother-infant approaches increased ($p < .05$) but were still very infrequent, rising from .12 in preseparation to .27 in reunion. All other mother-infant behavior showed nonsignificant increases, due mostly to transient increases on the first reunion day.

There were essentially no changes in in-

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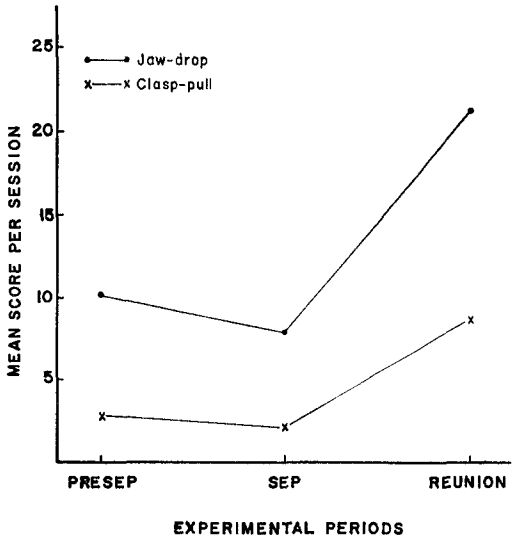


FIG. 3. Infant-infant close-quarters play behaviors.

fant-other mother, mother-other infant, mother-mother, and mother-nonsocial behaviors throughout the experiment.

Discussion

The initial reaction to maternal separation in patas infants paralleled that seen in the other primate species studied. This protest phase (Bowlby, 1960b) seemed to run its course more rapidly in the patas monkey, however. The depression phase usually seen following protest was both quantitatively and qualitatively different from that previously reported. The decline in play did occur; however, the patas infants remained alert. They attended to each other and the daily laboratory activities going on around them. Their motor activity was not severely depressed, as the high-home- and other-home-cage-location scores indicated. In general, the infants seemed to have suddenly grown much older. This impression is supported by the appearance during separation of the adult behaviors of aggression, thrusting, grooming, and just sitting watching the activities of the people and animals around them. Almost all of a mother patas's time is spent in the latter two activities.

Seay et al. (1970) indicated that patas infants maintain close contact with their mothers for a longer time than macaques. Whether patas infants would exhibit the more severe macaquelike reactions if separation were performed at an earlier age, perhaps at 4 months, would depend on whether the developmental difference, interacting with the greater permissiveness of patas mothers, produces earlier autonomy or instead fosters dependence.

The appearance of contact play and the increase in scores for the components of such play (clasp-pull, jaw drop) was surprising. Seay and Harlow (1965) noted a similar increase in rhesus infants, but contact play is the primary play pattern for that species, as noncontact play is for patas. They related the increase to heightened aggressiveness. The qualitative difference between aggression and contact play for the patas infants was clear-cut, however. The most parsimonious explanation may lie in the relation of the maternal role to peer interaction. The patas mother is protective of her infant when he is away from her. Maternal threats to the other infant are frequent, and are elicited most often when the infants stop chasing each other and begin to grapple. It would seem that the close contact which the infants experienced during separation fostered the development of close-quarters contact play. The males were typically bolder, attending less to the other mother's threats, but probably because two of the three pairs of infants were male-female, the mutual contact-play pattern was seen only in the male-male pair during reunion. While the patas mother inhibits contact-play behavior in infants, her presence promotes the primary noncontact-play pattern. When their mothers were returned, two infants approached them, briefly touched noses, then began playing with the other infant as they had before separation. The other subjects began more or less normal playing soon after return of the mother. The mere presence of the mother as a potential haven of safety was all that seemed to be needed to ameliorate the main effects of separation, namely, reduction in play behavior. The return of the mother interacted with the close infant-infant contact in separation to produce an increase in the primary noncontact-play pattern and a signifi-

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cant rise in the secondary contact-play pattern and its components.

Baker (1969) studied the effects of cross-species social interaction on patas and cynomologous (*Macaca irus*) mother-infant pairs. Each infant was permitted to interact with a mother-infant pair of the other species. The results indicated that maternal roles were modified and the development of peer socialization in both species was retarded. Attempts are now being made at the University primate laboratory to rear patas and macaque infants with mothers of different species in order to further clarify the variables involved.

In light of the differences among monkey species with regard to maternal separation, Bowlby's (1958) ethological theory of primary responses, and other theories based on instinctual responses, for example, Harlow (1961), may prove too limited to fully account for nonhuman data, as psychoanalysts already feel they are for human data (see Discussants in Bowlby, 1960a). Cairns's (1966) contiguity-learning approach and Gewirtz's (1961) instrumental learning analysis of mother-infant attachment may provide more flexible theoretical frameworks. However, no general theory can be successful unless species-specific variations in attachment behavior are taken into account.

REFERENCES

BAKER, R. P. The effects of interspecies infant interaction upon the social behavior of *Macaca irus* and *Erythrocebus patas*. Unpublished master's thesis, Louisiana State University, 1969.

BOWLBY, J. Maternal care and mental health. *World Health Organization Monograph Series*, 1952, No. 2.

BOWLBY, J. The nature of the child's tie to his mother. *International Journal of Psychoanalysis*, 1958, **39**, 350–373.

BOWLBY, J. Grief and mourning in infancy and early childhood. *Psychoanalytic Study of the Child*, 1960, **15**, 9–52. (a)

BOWLBY, J. Separation anxiety. *International Journal of Psychoanalysis*, 1960, **41**, 89–113. (b)

BOWLBY, J. Separation anxiety: A critical review of the literature. *Journal of Child Psychology and Psychiatry*, 1961, **1**, 251–269.

BOWLBY, J., AINSWORTH, M., BOSTON, M., & ROSENBLUTH, D. The effect of mother-child separation: A follow-up study. *British Journal of Medical Psychology*, 1956, **29**, 211–247.

CAIRNS, R. B. Attachment behavior of mammals. *Psychological Review*, 1966, **73**, 409–426.

CALDWELL, B. M., HERSHER, L., LIPTON, E. L., RICHMOND, J. B., STERN, G. A., EDDY, E., DRACHMAN, R., & ROTHMAN, A. Mother-infant interaction in monomatric and polymatric families. *American Journal of Orthopsychiatry*, 1963, **33**, 653–664.

GARDNER, D. G., HAWKES, G. R., & BURCHINAL, L. G. Noncontinuous mothering in infancy and development in later childhood. *Child Development*, 1961, **32**, 225–234.

GEWIRTZ, J. L. A learning analysis of the effects of normal stimulation, privation, and deprivation in the acquisition of social motivation and attachment. In B. M. Foss (Ed.), *Determinants of infant behavior*. Vol. 1. New York: Wiley, 1961.

GRIFFIN, G. A. *The effects of multiple mothering on the infant-mother and infant-infant affectional systems*. (Doctoral dissertation, University of Wisconsin) Ann Arbor, Mich.: University Microfilms, 1966. No. 66-13, 410.

HALL, K. R. L. Behavior and ecology of the wild patas monkey (*E. patas*) in Uganda. *Journal of Zoology: Proceedings of the Zoological Society of London*, 1966, **148**, 15–87.

HALL, K. R. L., & MAYER, B. Social interactions in a group of captive patas monkeys (*E. patas*). *Folia Primatologica*, 1967, **5**, 213–236.

HANSEN, E. W. The development of maternal and infant behavior in the rhesus monkey. (Doctoral dissertation, University of Wisconsin) Ann Arbor, Mich.: University Microfilms, 1962. No. 63-653.

HARLOW, H. F. The development of affectional patterns in infant monkeys. In B. M. Foss (Ed.), *Determinants of infant behavior*. Vol. 1. New York: Wiley, 1961.

HARLOW, H. F., & HARLOW, M. K. Social deprivation in monkeys. *Scientific American*, 1962, **207**(5), 136–146.

HARLOW, H. F., & ZIMMERMAN, R. R. Affectional responses in the infant monkey. *Science*, 1959, **130**, 421–432.

MITCHELL, G. D., HARLOW, H. F., GRIFFIN, G. A., & MØLLER, G. W. Repeated maternal separation in the monkey. *Psychonomic Science*, 1967, **8**, 197–198.

PINNEAU, S. R. The infantile disorders of hospitalism and anaclitic depression. *Psychological Bulletin*, 1955, **52**, 429–452.

ROSENBLUM, L. A., & KAUFMAN, I. C. Variations in infant development and response to maternal loss in monkeys. *American Journal of Orthopsychiatry*, 1968, **38**, 418–426.

SCHAFFER, H. R., & EMERSON, P. E. Patterns of response to physical contact in early human development. *Journal of Child Psychology and Psychiatry*, 1964, **5**, 1–14.

SEAY, B., HANSEN, E., & HARLOW, H. F. Mother-infant separation in monkeys. *Journal of Child Psychology and Psychiatry*, 1962, **3**, 123–132.

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SEAY, B., & HARLOW, H. F. Maternal separation in the rhesus monkey. *Journal of Nervous and Mental Diseases,* 1965, **140,** 434–441.

SEAY, B., SCHLOTTMANN, R. S., & THORNE, B. M. Maternal and filial behavior in monkeys. *Development Psychology,* 1970, **3,** 66–72.

SELYE, H., & FORTIER, C. Adaptive reaction to stress. *Psychosomatic Medicine,* 1950, **12,** 149–157.

SPENCER-BOOTH, Y., & HINDE, R. A. The effects of separating rhesus monkey infants from their mothers for six days. *Journal of Child Psychology and Psychiatry,* 1966, **7,** 179–197.

SPITZ, R. A. Hospitalism: An inquiry into the psychiatric conditions in early childhood. *Psychoanalytic Study of the Child,* 1945, **1,** 53–74.

SPITZ, R. A. Anaclitic depression. *Psychoanalytic Study of the Child,* 1946, **2,** 313.

SPITZ, R. A. Infantile depression and the general adaptation syndrome. In P. H. Hoch & J. Zubin (Eds.), *Depression.* New York: Grune & Stratton, 1954.

STOTT, D. H. Abnormal mothering as a cause of mental subnormality in the light of possible congenital factors. *Journal of Child Psychology and Psychiatry,* 1962, **3,** 79–91.

THORNE, B. M., SCHLOTTMANN, R. S., & SEAY, B. Single animal vs. pair observation in *Macaca irus. Journal of Genetic Psychology,* 1969, **115,** 17–32.

WINER, B. J. *Statistical principles in experimental design.* New York: McGraw-Hill, 1962.

YARROW, L. J. Separation from parents during early childhood. In M. L. Hoffman & L. W. Hoffman (Eds.), *Review of child development research.* Vol. 1. New York: Russell Sage Foundation, 1964.

(Received June 20, 1969)

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